Filipe Maschio • filipemaschio@yahoo.com

Network Security Incident Report:
ICMP Flood DoS Attack Analysis (NIST CSF)

Summary

1 Executive Summary 2
2. Organization Overview 2
3 Incident Description 2
Impact of the Incident 3
4 Incident Timeline 3
5 Attack Vector Analysis 4
6 Indicators of Compromise (IOCs) 4
7 NIST Cybersecurity Framework Analysis 5
Identify 5
Protect 5
Detect 5
Respond 6
Recover 7
8 Lessons Learned 7
9 Security Improvements Implemented 7
10 Conclusion 8

1 Executive Summary

On the morning of the incident, the organization experienced a network service disruption that prevented employees from accessing internal systems and network resources. Investigation revealed that the disruption was caused by a Denial-of-Service (DoS) attack involving a large volume of ICMP echo request packets directed at the company's network.

The malicious traffic overwhelmed the network infrastructure and prevented legitimate traffic from accessing internal services. The attack exploited a misconfigured firewall, which did not properly limit or filter incoming ICMP traffic.

The disruption lasted approximately two hours before the incident response team successfully mitigated the attack by blocking ICMP traffic and temporarily disabling non-critical services to reduce network load.

Following the incident, the cybersecurity team conducted an investigation and implemented additional security controls to prevent similar attacks in the future. This report analyzes the incident using the NIST Cybersecurity Framework (CSF) and outlines the improvements made to the organization's network security posture.

2. Organization Overview

The organization is a multimedia services company that provides digital solutions to small businesses, including:

- Web design services;
- Graphic design;
- Social media marketing management.

Because these services rely heavily on internet connectivity and internal network availability, maintaining a stable and secure network infrastructure is critical to daily business operations.

Any disruption to network availability can significantly impact productivity and service delivery to clients.

3 Incident Description

At approximately 09:15 AM, employees began reporting slow or unresponsive network services. Within minutes, internal systems such as shared drives, internal web tools, and network resources became inaccessible.

Initial network monitoring indicated an unusually high volume of incoming traffic targeting the network infrastructure.

Further analysis revealed that the incoming traffic consisted primarily of ICMP echo requests, commonly known as ping packets.

A malicious actor had launched a Denial-of-Service (DoS) attack by flooding the organization's network with ICMP packets, overwhelming available bandwidth and processing resources.

Because the firewall was not configured to limit ICMP traffic, the attack was able to reach internal systems and disrupt network operations.

The attack lasted approximately two hours before mitigation measures were successfully implemented.

Impact of the Incident

The ICMP flood attack caused significant disruption to the organization's network infrastructure.

Observed impacts included:

- Temporary unavailability of internal network services;
- High latency across internal and external network connections;
- Network device resource exhaustion due to excessive packet processing;
- Interruption of several non-critical services.

Although critical systems were eventually restored, the attack temporarily affected business operations and required immediate mitigation measures.

4 Incident Timeline

Time	Event
09:15	Employees report slow network performance
09:20	Network services become unresponsive
09:25	IT team detects abnormal traffic levels
09:30	Security team identifies ICMP flood attack
09:40	Firewall rules updated to block ICMP traffic

Time	Event
09:50	Non-critical services temporarily disabled
10:30	Network stability begins to recover
11:15	Critical services fully restored
11:30	Security investigation initiated

5 Attack Vector Analysis

The attack was identified as an ICMP Flood Denial-of-Service attack.

In this type of attack, a malicious actor sends a large number of ICMP echo request packets to a target system. These packets consume network bandwidth and processing resources, preventing legitimate traffic from reaching services.

Key characteristics of the attack included:

- Extremely high volume of ICMP packets;
- Saturation of network bandwidth;
- Increased CPU utilization on network devices;
- Disruption of legitimate internal network traffic.

The attack exploited a misconfigured firewall, which allowed unlimited ICMP packets into the network without filtering or rate limiting.

6 Indicators of Compromise (IOCs)

The investigation identified several indicators consistent with an ICMP flood Distributed Denial-of-Service (DDoS) attack.

Observed indicators included:

- Extremely high volume of ICMP Echo Request packets;
- Network bandwidth saturation during the attack window;
- Multiple source IP addresses sending large numbers of ICMP packets;
- Evidence of IP address spoofing attempts;
- Abnormal traffic spikes detected in network monitoring tools.

Network logs and firewall telemetry confirmed that the ICMP traffic exceeded normal baseline levels, indicating a coordinated flooding attempt intended to overwhelm network resources.

7 NIST Cybersecurity Framework Analysis

Identify

The incident response team conducted a review of network infrastructure and security configurations.

The investigation revealed several security gaps:

- Firewall misconfiguration allowing unrestricted ICMP traffic;
- Lack of network traffic rate limiting;
- Absence of advanced traffic monitoring tools;
- No intrusion detection or prevention system deployed.

The attack primarily affected the organization's internal network infrastructure, disrupting access to shared systems and internal services used by employees.

Protect

To strengthen the organization's security posture, several protective measures were implemented.

The network security team configured new firewall rules to limit the rate of incoming ICMP packets and block excessive traffic.

Additional security controls included:

- Source IP verification to detect spoofed addresses;
- Improved firewall configuration policies;
- Regular audits of firewall and network security settings;
- Development of improved network security procedures.

These safeguards help protect network resources from similar attacks in the future.

Detect

To improve threat detection capabilities, the organization implemented enhanced monitoring and detection mechanisms designed to identify abnormal traffic patterns and potential denial-of-service attacks in real time.

The following technologies and monitoring controls were deployed:

- Network traffic monitoring software capable of analyzing bandwidth usage, packet volume, and protocol distribution to identify unusual spikes in traffic, such as excessive ICMP requests.
- Intrusion Detection System (IDS) configured to inspect incoming network traffic and generate alerts when suspicious patterns are detected, including abnormal ICMP traffic volumes associated with DoS attacks.
- Intrusion Prevention System (IPS) integrated with the firewall to automatically block or filter malicious traffic based on predefined detection rules and behavioral analysis.
- Firewall log monitoring, enabling continuous analysis of connection attempts, protocol usage, and packet frequency to identify indicators of network abuse or malicious activity.

Additionally, the monitoring infrastructure was configured to trigger automated alerts when network activity exceeds predefined thresholds, such as:

- High volumes of ICMP packets within a short period of time;
- Sudden spikes in inbound network traffic;
- Repeated requests from suspicious or spoofed IP addresses.

These monitoring capabilities allow the security team to quickly identify anomalies, investigate potential threats, and respond to malicious activity before it significantly impacts network availability.

Respond

During the incident, the cybersecurity and IT teams worked together to contain and mitigate the attack.

Response actions included:

- Blocking incoming ICMP packets at the firewall;
- Temporarily disabling non-critical network services;
- Prioritizing restoration of critical services;
- Monitoring network traffic for additional malicious activity.

The incident was documented and communicated to IT leadership to support improvements in incident response procedures.

Recover

After the attack was mitigated, the organization focused on restoring normal operations.

Recovery actions included:

- Gradual restoration of network services;
- Continuous monitoring to confirm network stability;
- Implementation of new security controls;
- Documentation of lessons learned from the incident.

These recovery efforts ensured that the organization returned to normal operations while improving resilience against future cyber threats.

8 Lessons Learned

The incident highlighted several important areas for improvement in the organization's cybersecurity strategy.

Key lessons learned include:

- Firewall configurations must be regularly audited;
- Network traffic monitoring is critical for early detection;
- Rate limiting should be implemented for network protocols such as ICMP;
- IDS and IPS systems significantly improve threat detection and prevention.

Implementing these improvements strengthens the organization's ability to detect and mitigate future cyber attacks.

9 Security Improvements Implemented

Following the incident, the organization implemented several security improvements to reduce the risk of future DDoS attacks.

Implemented improvements include:

- Implementation of ICMP rate limiting rules on perimeter firewalls;
- Deployment of an Intrusion Detection and Prevention System (IDS/IPS) to identify abnormal traffic patterns;

- Improved network traffic monitoring and alerting thresholds;
- Configuration of source IP verification and anti-spoofing protections;
- Development of a DDoS response playbook for future incidents.

Additionally, the organization is evaluating the use of cloud-based DDoS mitigation services to provide additional protection against large-scale volumetric attacks.

10 Conclusion

The ICMP flood DoS attack demonstrated how a simple network misconfiguration can lead to significant service disruption.

By analyzing the incident using the NIST Cybersecurity Framework, the organization was able to identify weaknesses, implement stronger security controls, and improve its incident response capabilities.

Through continuous monitoring, improved firewall policies, and enhanced detection technologies, the organization is now better prepared to defend against similar attacks in the future.